# Chapter 1: Some Time Series Concepts – God is throwing Dice

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## 1 Overview

This chapter summarizes some basic concepts in times series analysis. The focus is on linear time series models, which will be the core tool in these notes. We will view the world through the prism of time series models. In particular, the economic models we consider will be cast in the form of time series.

We think of time series as repeated observations of some variable of interest, for example GDP, call it $x_t$. Our interest lies in characterizing the properties of this series. The main reason we are interested in doing so is to forecast it and understand how it responds to policy intervention. Thus we are interested in a joint distribution of a vector of the variable:

$$\mathbf{x} = [x_1, x_2, x_3, ....., x_T].$$

Theoretically, the vector $\mathbf{x}$ is a *single* observation of a stochastic variable. Obviously, it is difficult to draw any inference based on a single observation. Hence more assumptions are needed. This is the focus of time series analysis. We will typically refer to a time series such as $\mathbf{x}$ as $\{x_t\}$, where the curly bracket denotes that we are considering a sequence and $t$ is the time subscript.

Time series analysis is typically thought to have originated from an Econometrica article by Slutsky in 1937 (translated from a Russian article written 10 years earlier). Slutsky proposed the novel idea that we should think of $x_t$ as a result of some random shocks or what he called *incoherent* forces. A simple *time series model* of the evolution of $x_t$ is:

$$x_t = \mu_t + \rho x_{t-1} + \epsilon_t, \tag{1}$$

where $\rho$ is a parameter, $\epsilon_t$ is a random variable (for example generated by a coin toss, taking a value of +1 if heads and -1 if tails) and $\mu_t$ is a non-random

variable that is deterministic – or *coherent*, in Slutsky's original terminology.

The perspective that economic time series such as GDP are driven by equations like (1) was a significant leap in logic relative to how people perceived economics when the discipline started. Since Slutsky's philosophy underlies all modern macroeconomic models, often referred to as *dynamic stochastic general equilibrium (DSGE) models*, it deserves some general comments at the risk of sounding exceedingly vague.

Economics was modeled on physics. Adam Smith, in his book *The Wealth of Nations*, which is often thought of as having established economics as a discipline, claimed that his ambition was to discover *laws* of human affairs similar to those that Newton had proposed for physics. The reigning perspective of science at the time was that its role was to *explain* the evolution of a variable as a consequence of some well-understood fundamentals that could be measured. Laplace, in Stephen Hawking's paraphrasing, summarized this perspective as: "if at one time, we knew the positions and speeds of all the particles in the universe, then we could calculate their behavior at any other time, in the past or future." (Hawking, 2016) [2].

Clearly, equation (1) says otherwise. The value of $x_t$ is partially determined by a variable that can take on values that are randomly generated, e.g. via the flip of a coin. If equation (1) describes the world, then, as in quantum physics, economic variables can never be hoped to be fully explained and the evolution of GDP is partially generated by the randomness of chance. Einstein's famous objection to quantum physics was that "God does not throw dice." His criticism applies to modern economics.

Despite Einstein's protest, physics moved from determinism towards embracing that the universe will, to some extent, evolve in arbitrary, unforecastable ways. Within the economic profession, however, the notion that unexplainable *shocks* are driving the business cycle is still highly controversial. Criticism of the modern macroeconomic models that will be the core of this class, known as Dynamic Stochastic General Equilibrium (DSGE) models, has been especially prominent since the financial crisis of 2008, with two major arguments.

First, these models are claimed to be useless because they did not predict the crisis. The Queen of England, for example, famously asked economists at the London School of Economics, "Why did no one see this coming?" Second, these models are criticized for having "implausible shocks" that cannot be easily

explained or directly measured by the data (see for example Romer (2016) who refers to them dismissively as "imaginary" and "post-real"). According to the Slutsky philosophy, however, some part of the variance of economic variables is inherently "incoherent" or unforecastable. God is throwing dice. Hence, if one adapts that philosophy, there is little hope the criticism of the Queen of England will ever be fully addressed by economic models, as some part of the evolution of variables is inevitably going to be "incoherent." Moreover, it is not clear that the underlying stochastic disturbance can ever be precisely identified. In Slutsky's terminonlogy the shocks are simply "incoherent." Thus, it is entirely possible that they can never be precisely understood.[1]

A separate criticism of modern DSGE models, all of which essentially boil down to equations such as (1), is related to the *nature* of the estimated shocks. This line of criticism is exemplified by Chari, Kehoe and McGrattan (2008) [1]. Their argument is that modern estimated models have shocks that are *not structural*. What they mean by not structural is that the value of the shock will *change* in response to different policies. This is a major concern. It seems unavoidable that all economic models will always have some elements that will be due to chance. If the assumed random generating process, however, depends strongly on what kind of policy is followed, then it is impossible to use the model to evaluate different policies.

Our principal focus will be on models of a form analogous to (1), although they will involve several variables, with past values as well as expected future values of variables entering the equations. However, they will all be *linear* stochastic difference equations. The assumption of linearity is a major limitation. The reason for accepting this limitation is the large gain in tractability it buys. To put macro models, which are typically non-linear, into form (1) requires linearizing the models, as explained in Chapter 3. This means that the analysis will only be valid locally, i.e. as long as the shocks are not "too large." We will only briefly comment on methods exploring non-linearities.

# 2 Time Series Terminology

This section mostly defines language and terminology that will play an important role in later chapters. My goal is to expose you to notation, language and terms, rather than detailed derivation of results.

[1] Consistent with this, recent experimental evidence suggests that even simple and exactly-the-same decisions *of the same individual in a controlled environment* have stochastic or "incoherent" elements, see Woodford (2014).

**Unconditional mean**

The expectation of the $t^{th}$ observation of a time series refers to the mean of this probability distribution, called the unconditional mean of $Y_t$:

$$E(Y_t) = \mu_t$$

where the variable $E$ denotes expectations. Example: suppose $\{Y_t\}$ is generated by coin tosses. The expected value of a coin toss with a payoff of -1 for tails and +1 for heads is: $E(Y_t) = \left(\frac{1}{2} \cdot -1\right) + \left(\frac{1}{2} \cdot 1\right)$.

**Variance**

The variance of the random variable $Y_t$ (denoted $\gamma_{0t}$) is:

$$\gamma_{0t} = E(Y_t - \mu_t)^2$$

**Autocovariance**

The autocovariance of $Y_t$ is the covariance of $Y_t$ with its own $j^{th}$ lagged value:

$$\gamma_{jt} = E(Y_t - \mu_t)(Y_{t-j} - \mu_{t-j})$$

**Covariance Stationary**

A process is said to be covariance stationary if:

i. $E(Y_t) = \mu$ for all $t$;

ii. $E(Y_t - \mu_t)(Y_{t-j} - \mu_{t-j}) = \gamma_j$ for all $t$.

This means that neither the mean nor the autocovariances depend on date $t$. Several variables of interest are not covariance stationary. For example, GDP clearly increases with a trend, as does the price level.

If output is detrended, however, one can plausibly claim that the *detrended* output is stationary. Similarly, the first difference of the log of the price level, i.e. *inflation*, can plausibly be claimed to be covariance stationary.

The property of covariance stationarity is important, and typically the time series we will work with are assumed to be covariance stationary. In many cases this means that the variables need to be detrended.

**White Noise**

A white noise is a sequence $\{\epsilon_t\}$ whose elements have mean zero and variance $\sigma^2$, and for which the $\epsilon_t$*'s* are uncorrelated across time:

i. $E(\epsilon_t) = 0$;

ii. $E(\epsilon_t^2) = \sigma^2$;

iii. $E(\epsilon_t \epsilon_{t+j}) = 0$ for all $j \neq 0$.

Usually we assume normally distributed white noise, i.e. $\epsilon_t \sim N(0, \sigma^2)$.

**AR(p), MA(p) and ARMA(p,q)**

A *moving average process* of order $q$, MA(q), is a weighted average of the $q$ most recent values of a white noise defined as:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + ... + \theta_q \epsilon_{t-q}$$

where $\{\theta_1, ..., \theta_q\}$ can be any real numbers.

An *autoregressive process* of order $p$, AR(p), is given by $Y_t$ as a linear combination of $p$ lags of itself and one white noise process:

$$Y_t = \mu + \phi_1 Y_{t-1} + .... + \phi_p Y_{t-p} + \epsilon_t$$

An *autogressive moving average process* of order $(p, q)$, ARMA(p,q), combines an AR(p) and MA(q) process, i.e.:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + ... + \theta_q \epsilon_{t-q} + \phi_1 Y_{t-1} + ... + \phi_p Y_{t-p}$$

ARMA(p,q) can be vector-valued models. The most important example is Vector Autoregression (VAR).

**Vector Autoregression (VAR)***

A vector autoregression is a vector-valued AR(p) model:

$$Z_t = A_1 Z_{t-1} + A_2 Z_{t-2} + .... + A_q Z_{t-p} + v_t$$

where $Z_t$ is a vector of variables of dimension $m$ (for example output, inflation, interest rate) and $v_t$ is a vector of the forecast errors of $Z_t$. In practice, an econometrician typically includes enough lags (p) so that the forecast errors $v_t$ are not correlated across time.

The forecast errors at a time $t$ $(v_{1t}, v_{2t}, ...., v_{mt})$, however, are typically correlated with each other *at a given time t* (thus, for example, the forecast error of inflation is typically correlated to the forecast error of output). Additional assumptions are needed to identify "structural shocks," that is, shocks that are independent from one another.

The most simple such procedure is the Cholesky decomposition, which supposes that the first forecast error, $v_{1t}$, is a structural shock, and then the part

of $v_{2t}$ that is not correlated to $v_{1t}$ is another structural shock, and then the part of $v_{3t}$ not correlated with $v_{1t}$ or $v_{2t}$ is another and so on (Note that when the residuals are defined in this way, we can find the structural shocks via simple OLS, i.e. the first structural shock is just the forecast error $v_{1t}$ and to find the second just regress $v_{2t}$ on $v_{1t}$ and the residual is the second structural shock, and so on).

# 3 Conditional Expectations and the Law of Iterated Expectations

**Conditional Expectations**

Before we defined the unconditional expectation of the time series $Y_t$, thus considering all possible contingencies of the time series. The conditional expectation of $Y_t$ at time $t-1$ is denoted as $E_{t-1}$. Consider, for example, the process:

$$Y_t = \rho Y_{t-1} + \epsilon_t$$

where $\epsilon$ is a zero mean iid. Then the conditional expectation of $Y_t$ at time t-1 is given by:

$$E_{t-1}Y_t = \rho E_{t-1}Y_{t-1} + E_{t-1}\epsilon_t = \rho Y_{t-1}$$

Note that we will typically assume that the variables that have the time index $t$ are known at time $t$. Thus $E_t Y_t = Y_t$. This notation deserves a footnote.[2]

**Law of Iterated Expectations (LIE)**

Consider $t+j \geq s \geq t$. The Law of Iterated Expectations states that:

$$E_t E_s Y_{t+j} = E_t Y_{t+j}$$

This is proved in many econometric text books. We will use the $LIE$ whenever we solve a model "forward" as we will see shortly.

# 4 Impulse Response Functions

Consider a stochastic process $\{Y_t\}$ that is a function of the sequence $\{\epsilon_t\}$. The impulse response function of $\{Y_t\}$ with respect to $\epsilon_t$ is the partial derivative $\frac{\partial Y_{t+j}}{\partial \epsilon_t}$. For example, consider the the process:

$$Y_t = \rho Y_{t-1} + \epsilon_t$$

[2]This notation is not universal. For example, it is very common that people denote the capital stock determined at time $t$ by $K_{t+1}$. This implies that $E_t K_{t+1} = K_{t+1}$. While I believe this is a prime example of very bad notation, the profession appears not to agree with me on that point. I will even violate my own principles in some later chapters.

The impulse response function of $Y_t$ with respect to $\epsilon_t$ is given by:

$$\frac{\partial Y_{t+j}}{\partial \epsilon_t} = \rho^j.$$

To see this, bring forward the AR process once and substitute in for $Y_t$, i.e. $Y_{t+1} = \rho Y_t + \epsilon_{t+1} = \rho(\rho Y_{t-1} + \epsilon_t) + \epsilon_{t+1} = \rho^2 Y_{t-1} + \rho\epsilon_t + \epsilon_{t+1}$, so that $\frac{\partial Y_{t+1}}{\partial \epsilon_t} = \rho$. You can similarly solve for $Y_{t+2}$ using the following solution: $Y_{t+2} = \rho Y_{t+1} + \epsilon_{t+2} = \rho(\rho^2 Y_{t-1} + \rho\epsilon_t + \epsilon_{t+1}) + \epsilon_{t+2} = \rho^3 Y_{t-1} + \rho^2\epsilon_t + \rho\epsilon_{t+1} + \epsilon_{t+2}$, so that $\frac{\partial Y_{t+2}}{\partial \epsilon_t} = \rho^2$. Repeating this again for $Y_{t+3}$ and any $Y_{t+j}$, the result is obtained.

# 5 Representation Theorem of Time Series: The Wold Theorem*

We have seen several examples of time series models, for example AR(p) and MA(q). How different are these models? Consider the AR(1):

$$Y_t = \rho Y_{t-1} + \epsilon_t. \tag{2}$$

Since the above equation holds at $t$, it also holds at $t-1$, giving $Y_{t-1} = \rho Y_{t-2} + \epsilon_{t-1}$. Substitute this into the original equation to obtain: $Y_t = \rho(\rho Y_{t-2} + \epsilon_{t-1}) + \epsilon_t = \rho^2 Y_{t-2} + \rho\epsilon_{t-1} + \epsilon_t$. The same equation also holds at time $t-2$, i.e $Y_{t-2} = \rho Y_{t-3} + \epsilon_{t-2}$. Again, substitute into the last equation to obtain: $Y_t = \rho^3 Y_{t-3} + \rho^2\epsilon_{t-2} + \rho\epsilon_{t-1} + \epsilon_t$. Repeat this an infinite number of times to obtain, as long as $\rho < 1$ :

$$Y_t = \sum_{s=0}^{\infty} \rho^s \epsilon_{t-s}. \tag{3}$$

Hence, an AR(1) can be expressed as a **MA($\infty$) − the AR(1) process in (2) and the MA($\infty$) in (3) are different representations of one and the same underlying stochastic process.**

- If $\epsilon$ is normal, then an ARMA process is fully characterized by first and second moments (mean, variance, covariance).
- If two Gaussian ARMA processes have the same autocovariance functions, then they are the same process (the probability of any event happening under one process is the same as under the other).

We next state (informally) an important result:

**Proposition 1** (Wold Representation)**.** *All covariance-stationary time series processes can be represented by/be decomposed into a deterministic component and a* $MA(\infty)$*.*

We do not attempt to prove this proposition.[3] This result is often cited as a foundation for VAR.

# 6 References


[1] V. V. Chari, Patrick J. Kehoe, and Ellen R. McGrattan. "New Keynesian Models: Not Yet Useful for Policy Analysis". *American Economic Journal: Macroeconomics* 1.1 (2009), pp. 242–266.

[2] Stephen Hawking. "Does God Play Dice?" (2016). URL: `http://www.hawking.org.uk/does-god-play-dice.html`.

[3] Paul Romer. "The Trouble With Macroeconomics" (2008). URL: `http://paulromer.net/wp-content/uploads/2016/09/WP-Trouble.pdf`.

[4] Eugen Slutzky. "Stochastic Choice: An Optimazing Neuroeconomic Model". *Econometrica* 5.2 (1937), pp. 105–146.

[5] Michael Woodford. "Stochastic Choice: An Optimazing Neuroeconomic Model". *American Economic Review* 104.5 (2014), pp. 495–500.


# 7 Problems

**Problem 1**
Consider the following time series model:

$$x_t = \rho x_{t-1} + \epsilon_t, \tag{4}$$

where $\epsilon_t$ is an iid and can take on value -0.5 with probability 0.5 and +0.5 with probability 0.5.

i. Under what condition is this process covariance stationary? Show that it is not covariance stationary if $\rho = 2$.

ii. Let us assume that $\rho = 0.75$. By flipping a coin please generate 15 datapoints (or more if you want) for this time series process, assuming $head = -0.5$ and $tail = 0.5$. Report it and plot up the sample path from this experiment with $x_t$ as a function of time.

[3] A nice proof and discussion is found on Larry Christiano's homepage, see http://faculty.wcas.northwestern.edu/~lchrist/finc520/wold.pdf.

iii. Now assume that you are the econometrician and only see the time path you just plotted up for $x_t$. You know the form of the stochastic process given in equation (4) but you don't know the exact value of $\rho$. Please estimate $\rho$, denoted by $\hat{\rho}$.

***Hint:*** *You can use the formula:*

$$\hat{\rho} = \frac{\sum_{t=1}^{15} x_t x_{t-1}}{\sum_{t=1}^{t=15} x_t^2}$$

*and use 0 as an initial value for* $x_0$.

iv. Plot on the same graph the actual impulse response function and your estimated one (i.e. use both $\rho$ and $\hat{\rho}$).

v. Compute your forecast for $x_{16}$ using in one case $\rho$ and in the other case $\hat{\rho}$. Now consider your forecast for $x_{t+j}$ as $j$ becomes very big, both for $\rho$ and your estimated value for $\hat{\rho}$. How different is your long-run forecast depending on whether you use $\rho$ or $\hat{\rho}$?

vi. Suppose you toss your coin for the $16^{th}$ time and get head. How does this affect your forecast for $x_t$ in 10 periods (i.e. after you have tossed your coin for 10 more time)?

**Problem 2**

Please represent the $AR(1)$ process above as a $MA(\infty)$. What is the condition on $\rho$ for this representation to be valid?

**Problem 3**

Consider the following stochastic process:

$$x_t = 0.5\epsilon_t + 1.5\epsilon_{t-1},$$

where $\epsilon_t$ is iid N(0,1).

i. Is this process covariance stationary?

ii. Plot up the impulse response function for this process.

iii. What is the autocovariance function, i.e., $\gamma_x(j)$ for $j$=1,2,3,4....?